H-Stacked Multiverse

A Relational Model of Layered Worlds, Subtraction, and Recognition

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Abstract

This paper proposes the H-Stacked Multiverse, a speculative framework in which reality is represented as a hierarchy of structurally related world-states $H_0, H_1, \ldots, H_n$. Each layer possesses its own relational grammar: a set of states, relations, representations, recognition rules, and transformations. A transition between layers is not necessarily physical travel between universes; it can instead be a transformation of the structure that determines what constitutes a world. In particular, subtraction may generate a new layer when removing a relation or component changes the identity conditions of the remaining structure. The model is presented as a mathematical-philosophical framework, not as experimental evidence for physically existing parallel universes.

Keywords: multiverse, H-stack, relational structure, subtraction, recognition, world-model, matrix, ontology, computational perception

1. Introduction

The H-Stacked Multiverse develops the intuition that a world is not merely a collection of objects. It is also a structure specifying how objects relate, how distinctions are maintained, and how entities become recognizable. The framework represents a world as

$$H = (X, R, \Lambda, \Gamma, T)$$

where X denotes states or entities, R their relations, Λ the representational or linguistic system, Γ the recognition rule, and T the set of permitted transformations. A universe is consequently treated as a self-consistent relational structure rather than only as a container of matter.

2. The H-Stack

The multiverse is represented as a sequence of layers:

$$\mathbb{H} = \{H_0, H_1, H_2, \ldots, H_n\}$$

The index n is interpreted primarily as structural depth rather than ordinary physical time. Each layer may be internally coherent while remaining related to other layers through transformation maps.

$$H_n \to^{\Phi_n} H_{n+1}$$

The resulting structure need not be a simple tree. If multiple transformations are possible, the H-stack becomes a directed network of world-generating states.

3. Subtraction as a World-Generating Operation

Let a structural component S be removed from a world H. A naive description would be H − S. However, if S participates in the definition of other entities, then removing it can alter the recognition rule itself:

$$\Gamma(H - S) \neq \Gamma(H)$$

The remainder is therefore not necessarily equivalent to the original world. A normalized successor layer can be written as:

$$H_{n+1} = N(H_n - S_n)$$

Here N represents the reorganization required to obtain a coherent successor structure. Subtraction consequently becomes potentially generative: removing structure can change the grammar of what remains.

4. Identity Across Layers

An entity may persist, transform, or dissolve across the H-stack. With a recognition function Γ_n, persistence occurs when

$$\Gamma_n(x) = \Gamma_{n+1}(\Phi_n(x))$$

while structural transformation occurs when the corresponding recognized identity changes. An entity may also have no recognized counterpart in the successor layer. This distinguishes physical or informational continuity from continuity of identity.

5. H-Distance

A structural distance between layers may be decomposed as

$$D = \alpha D_r + \beta D\Lambda + \gamma D\Gamma + \delta D_t$$

where the terms measure changes in relations, representation, recognition, and transformation rules. Two layers may therefore contain similar objects while remaining structurally distant if their recognition rules differ.

6. Recognition Boundaries

For a given layer define the recognizable domain

$$R_n = \{x \mid \Gamma_n(x) \text{ is defined}\}$$

An entity outside R_n need not be impossible. It is simply not represented as an identified entity under that layer's grammar. A transition may consequently produce a structure that becomes recognizable only at a higher layer:

$$x \notin R_n,\ x \in R_{n+1}$$

7. Language as Projection

Language or representation can be treated as a projection from world structure:

$$\Pi_n : H_n \to \Lambda_n$$

Because projection can discard distinctions, a transformed world may require a different vocabulary or representation. The H-stack therefore permits world transitions in which the old descriptive system no longer provides a complete representation of the new structure.

8. Observer-Dependent H-Stacks

An observer can be represented by a recognition function Γ_k. The resulting experienced or computationally represented world is

$$H^{(k)} = (X, R, \Lambda, \Gamma_k, T)$$

Different observers can therefore construct different world-models from the same underlying relational structure. This introduces a distinction between ontological multiplicity, structural multiplicity, and epistemic multiplicity.

9. Matrix Representation

For computational experimentation, a layer can be encoded by an adjacency or influence matrix A_n. A transition can be represented by

$$A_{n+1} = P_n A_n P_n^{-1} + \Delta_n$$

where P_n preserves selected relational structure and Δ_n represents structural change. A subtraction-based transformation may instead be written

$$A_{n+1} = N(A_n - S_n)$$

This provides a practical route for implementing H-stacks as graph or matrix transformation systems.

10. H-Stack Recursion

A general recursive formulation is

$$H_{n+1} = F(H_n, \Phi_n, S_n, \Gamma_n)$$

Repeated application generates a hierarchy of structurally transformed worlds. If H_{n+k} is isomorphic to H_n, the system exhibits structural recurrence. If it approaches but never reaches an equivalent state, the system instead approaches a structural attractor.

11. Equifactima and Cross-Layer Recognition

An equifactima can be treated within this framework as a relational equivalence recognized across distinct layers. For $x \in H_i$ and $y \in H_j$, define $x \sim_h y$ when a chain of H-transformations preserves the relevant relational structure. The objects need not be identical; their equivalence arises from structural correspondence.

12. The H-Stack Principle

The central principle of the framework can be stated as follows:

A world is a self-consistent relational structure, and a new world-layer arises when a transformation changes its relations or recognition rules sufficiently that the previous world-description is no longer equivalent.

$$H_{n+1} = N(H_n - \Delta_n)$$

Under this interpretation, subtraction is not necessarily mere destruction. It can be a mechanism through which new structural organization appears.

13. Physical Interpretation and Limits

The H-Stacked Multiverse should not be presented as empirical confirmation that parallel physical universes exist. Its immediate status is mathematical and philosophical. A physical interpretation would require a separate theory connecting the abstract objects H_n, transformation operators, and distances to measurable physical quantities.

14. Conclusion

The H-Stacked Multiverse treats a universe as more than a collection of contents. It is a relational grammar that determines how states, identities, and distinctions can be recognized. Transformations between layers may preserve some structure while altering other components. In the limiting case, a subtractive operation can change the recognition conditions so substantially that the successor is best regarded as a new world-layer.

The resulting conceptual multiverse is therefore a stack of possible relational grammars. Its central question is not simply how many universes exist, but what transformation makes one world cease to be structurally equivalent to another.

Conceptual note: This paper is a speculative mathematical-philosophical model and does not establish the physical existence of a multiverse.